THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · EASY LEVEL UNIT TEST SET 3

Chapters 1–5: **A Square and a Cube, Power Play, A Story of Numbers, Quadrilaterals, Number Play** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which of the following is a perfect square? 1

(a) 61 **(b)** 70 **(c)** 64 **(d)** 66

2. $3^2 \times 3^5$ equals: 1

(a) 3^7 **(b)** 3^4 **(c)** 9^7 **(d)** 3^{10}

3. The Roman symbol for 50 is: 1

(a) V **(b)** X **(c)** L **(d)** I

4. The diagonals AC and BD of a parallelogram meet at O. If AO = 4 cm, then AC is: 1

(a) 8 cm **(b)** 6 cm **(c)** 4 cm **(d)** 12 cm

5. Which of the following is a palindrome? 1

(a) 4554 **(b)** 4564 **(c)** 4543 **(d)** 4555

6. The value of 14^2 is: 1

(a) 28 **(b)** 182 **(c)** 196 **(d)** 210

7. $3^8 \div 3^3$ equals: 1

(a) 3^{11} **(b)** 3^6 **(c)** 3^5 **(d)** 3^{24}

8. The value of the Roman numeral XXIV is: 1

(a) 34 **(b)** 26 **(c)** 24 **(d)** 22

9. A quadrilateral has: 1

(a) 4 sides and 3 vertices **(b)** 5 sides and 5 vertices
(c) 3 sides and 3 vertices **(d)** 4 sides and 4 vertices

10. The LCM of 4 and 6 is: 1

(a) 24 **(b)** 16 **(c)** 3 **(d)** 12

11. $\sqrt{289}$ is equal to: 1

(a) 18 **(b)** 19 **(c)** 17 **(d)** 16

12. $(3^2)^4$ equals: 1

(a) 3^9 **(b)** 3^8 **(c)** 3^{16} **(d)** 3^6

13. The Roman numeral for 19 is: 1

(a) XIX **(b)** XVIII **(c)** XX **(d)** XXIX

14. Three angles of a quadrilateral are 65°, 85° and 110°. The fourth angle is: 1

(a) 90° **(b)** 110° **(c)** 80° **(d)** 100°

15. Is 5471 divisible by 3? 1

(a) Yes **(b)** Only if the number is a perfect square

(c) Only if the last digit is even **(d)** No

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the value of 13^3. 2

17. Express 28000 in scientific notation. 2

18. Write 7 in binary. 2

19. Name the quadrilateral whose four sides are equal and four angles are right angles. Give one more property. 2

20. Find the LCM of 3 and 4 by listing multiples. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the HCF and LCM of 12 and 16. 3

22. Find $\sqrt{324}$ by prime factorisation. 3

23. Simplify $3^4 \times 3^3 \div 3^5$ and find its value. 3

24. Write XXXVIII and CCXXIX as Hindu-Arabic numbers. 3

OR

In a parallelogram ABCD, $\angle A = 78^\circ$. Find $\angle B$, $\angle C$ and $\angle D$. 3

25. Test whether 2865 is divisible by 2, 5 and 10. 3

OR

Find $\sqrt[3]{729}$ by prime factorisation. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. ABCD is a rhombus of side 7 cm with $\angle A = 66^\circ$. Find $\angle B$, $\angle C$, $\angle D$ and the perimeter. 5

OR

Find the smallest number that must be added to 4537 so that it becomes divisible by 9. 5

27. Find the smallest number by which 252 must be multiplied so that the product is a perfect square. Also find the square root of the product. 5

OR

Find the values of 5

(i) 3^3,

(ii) 2^{-4},

(iii) 9^0 and

(iv) the sum of the three values.

END OF QUESTION PAPER — Total: 50 Marks